
Ontology-Guided Query Coverage Bias for Multi-Tool Selection

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Multi-tool agent prompts require the model to emit more than one tool in sequence,
2 so the central problem is coverage rather than single-step ranking. We evaluate an
3 ontology-guided query-space bias that suppresses ontology-aligned query compo-
4 nents at q_{proj} and compare it against a stationary key-side bias on the same
5 ontology basis. On MetaTool Subtask4 with 497 two-tool queries, the query-side
6 bias improves macro F1 from 0.7307 to 0.7471 on Qwen2.5-7B-Instruct and from
7 0.6227 to 0.6271 on Llama-3.1-8B-Instruct, whereas the stationary key-side bias is
8 negative on both models, dropping to 0.6850 on Qwen and 0.3105 on Llama. The
9 result is basis-specific rather than a generic low-rank effect: on Qwen, replacing
10 the ontology basis with a feature-shuffled or random control reduces the query-side
11 result to 0.7254 and 0.7068, while the corresponding key-side controls collapse to
12 0.0000. Additional diagnostics show that the useful region is narrow, with the best
13 Q-only point at $\beta = -0.1$, and that the real ontology perturbation is larger than
14 the null controls in effective attention magnitude, ruling out the explanation that
15 it succeeds merely because it perturbs less. The paper’s claim is therefore narrow
16 but coherent: for multi-tool selection, query-side coverage bias is the productive
17 direction, stationary key-side amplification is not, and the ontology basis matters
18 because it identifies a specific stable subspace rather than an arbitrary projection
19 family.

20 1 Introduction

21 Tool-using language models are often evaluated as if each query required a single decision. That
22 assumption is too weak for many agent settings. A user may simultaneously ask for news and market
23 impact, or for travel planning and exchange-rate information. In such cases the model must emit
24 multiple tools in sequence, not merely rank a single option.

25 This paper studies that multi-tool regime. Our starting point is a simple failure mode: a stationary
26 attention intervention that boosts one facet at every decoding step tends to keep boosting the same
27 facet after the first tool has already been emitted. On a multi-tool query, this is the wrong inductive
28 bias. The model needs a mechanism that makes the second step less like the first one.

29 The intervention we evaluate is an ontology-guided query bias defined on the same per-head basis
30 used by our key-side steering experiments. Its practical implementation is deliberately simple: at each
31 layer, we project the query onto the ontology subspace and add a signed multiple of that projection.
32 Negative coefficients suppress ontology-aligned query components and act as a coverage bias. The
33 hypothesis is direct: for multi-tool prompts, this suppression should improve second-tool recovery,
34 whereas a stationary key-side bias on the same basis should either do nothing or over-concentrate on
35 the first facet.

36 The evidence supports that hypothesis in a specific and interpretable way. On MetaTool Subtask4,
37 the Q-side bias is consistently positive on both Qwen and Llama, while the stationary K-side bias is
38 negative on both and catastrophic on Llama. The most important pattern is therefore not raw gain
39 magnitude but sign consistency, decisive null controls, and basis specificity.

The paper makes three contributions. First, it identifies a task mismatch in the closest prior line: stationary key-side amplification is a plausible design for prompt highlighting, but it is poorly matched to sequential coverage when the model must emit more than one tool. Second, it studies a deliberately simpler alternative, a training-free ontology-guided query-space suppression operator, and shows that its sign is consistently positive while stationary key-side amplification is negative on the same multi-tool benchmark. Third, it shows that the effect is basis-specific rather than a generic low-rank perturbation and places the perturbation analysis on a rigorous diagnostic footing. The central conclusion is therefore narrow but defensible: once the task requires sequential recovery of more than one tool, intervention site and sign matter more than the generic fact that the operator is low-rank.

2 Related Work

The closest prior line is spectral key-space steering. SEKA learns low-rank projections from synthetic question-answer pairs and injects them into attention keys during inference; AdaSEKA extends that recipe with a small expert ensemble and an input-dependent expert selector [Li et al., 2026]. That family belongs to the broader inference-time steering literature, which also includes activation-space direction methods such as ITI and CAA [Li et al., 2023, Rinsky et al., 2024], spectral activation editing [Qiu et al., 2025], post-softmax attention steering [Zhang et al., 2024], prompt-anchoring methods [Tian and Zhang, 2025], and key/query focus-direction methods for long-context relevance control [Zhu et al., 2025]. The common theme is intervention without finetuning, but the intended effect is usually to preserve, amplify, or recover a chosen signal.

Our paper differs first in the *task*. We study multi-tool selection, where the model must emit two tools in sequence on the same prompt. This changes the failure mode. A method that keeps amplifying the same facet can look reasonable on a single-selection benchmark and still be mismatched for sequential coverage, because the second decoding step should become less like the first one, not more. The relevant comparison to SEKA and AdaSEKA is therefore not “which low-rank edit is bigger,” but “which intervention matches the structure of sequential coverage.”

The paper also differs in the *intervention site and direction source*. The evaluated positive result comes from a query-side operator, not from key amplification, and the basis is derived from the tool catalog rather than learned from synthetic contrastive pairs. This distinction matters. AdaSEKA makes the selected key-side projector input-dependent, but the operation is still to amplify the chosen subspace. ITI and CAA edit activations along learned behavior directions, but they are not formulated as multi-tool coverage operators. Focus Directions changes key/query behavior to increase attention to relevant context, but its objective is long-context relevance rather than sequential tool diversity. Our method changes the sign semantics themselves: a negative query-space coefficient suppresses already dominant ontology-aligned components and therefore acts like a coarse coverage prior rather than a highlighting mechanism.

More broadly, the paper belongs to post-hoc activation steering rather than training-time routing. The intervention is applied at inference time through attention-space hooks, with no additional finetuning and no external selector network. That choice is deliberate. It lets the experiment isolate a narrow claim that can be cleanly tested against SEKA-style methods: before adding explicit history tracking or learned routing, does the combination of ontology geometry, intervention site, and sign already determine whether the effect encourages coverage or repetition?

KV-cache compression is orthogonal to the main question here. Methods such as KIVI study low-bit cache representations and memory-quality tradeoffs rather than tool-selection dynamics [Liu et al., 2024]. The evaluated models are instruction-tuned Qwen, Llama, and Mistral variants [Qwen Team et al., 2024, Grattafiori et al., 2024, Jiang et al., 2023], and the main benchmark is MetaTool Subtask4 because it directly exposes the sequential-coverage problem that stationary single-facet steering is expected to mishandle.

3 Method

3.1 Ontology basis

Each experiment uses a per-head ontology basis $B_{\text{ont}} \in \mathbb{R}^{L \times H_{\text{kv}} \times d \times r}$ constructed from the tool catalog. Intuitively, the basis spans directions that are repeatedly used by tool facets such as finance, news, travel, or media. The basis is fixed at evaluation time and shared by both the Q-side and K-side interventions.

Table 1: Proof-status map for the main claims.

Claim	Type	Fully proved?	Depends on modeling assumptions?
Stationary key-side steering does not introduce an explicit coverage variable	Structural	Yes	Minimal
Negative query-side projection has the correct sign semantics for coverage	Structural	Yes	Minimal
Attention output perturbation is controlled by qaMSE and a ρ^4 remainder	Diagnostic	Yes	Boundedness assumptions
Q-bias improves second-tool recovery on MetaTool Subtask4	Empirical	No	Must be shown experimentally

94 3.2 Practical query-space coverage bias

95 For a query tensor q , the implemented Q-side intervention is

$$q' = q + \beta B_{\text{ont}} B_{\text{ont}}^\top q.$$

96 When $\beta < 0$, the operator subtracts the query component aligned with the ontology subspace. In code
 97 this is implemented as a forward hook on `q_proj`. The method is history-free: it does not explicitly
 98 track which tool has already been emitted. We therefore interpret it as a practical approximation to a
 99 stronger ideal coverage operator, not as literal per-step facet-state tracking.

100 3.3 Stationary key-side bias

101 The key-side comparison uses the same basis but a different site:

$$k' = k + \alpha B_{\text{ont}} B_{\text{ont}}^\top k.$$

102 This intervention is stationary by construction. It amplifies the same ontology-aligned key directions
 103 at every decoding step. If the first emitted tool already occupies the dominant facet, the operator has
 104 no mechanism to suppress it at later steps.

105 3.4 Working hypothesis

106 The paper’s working hypothesis has two parts. The first is an *accuracy hypothesis*: negative query-
 107 side ontology bias should improve multi-tool recovery because it reduces repeated use of the same
 108 dominant facet. The second is a *specificity hypothesis*: if the ontology basis is replaced by a random
 109 or feature-shuffled basis of the same rank, the effect should disappear. The experiments are designed
 110 to test these two claims directly.

111 4 Theory

112 The paper needs only a compact formal package. The first two statements are structural: they ex-
 113 plain why stationary key-side amplification is mismatched to sequential coverage and why query-side
 114 suppression is the correct history-free surrogate. The theorem then places the perturbation language
 115 on a rigorous footing.

116 **Scope.** The theory is intentionally narrow. It proves two structural facts about history-free inter-
 117 ventions and one diagnostic perturbation bound. It does *not* prove benchmark F1 directly, and it does
 118 not claim exact facet-history tracking.

119 **Standing assumptions for Theorem 1.** Only the perturbation theorem needs explicit analytical
 120 assumptions. We use:

- 121 1. bounded query norm $\|q\| \leq Q_{\text{max}}$;
- 122 2. bounded value norm $\|v_t\| \leq V_{\text{max}}$;
- 123 3. row-wise key perturbation bound $\|e_t\| \leq \rho$.

124 The propositions do not rely on these boundedness assumptions.

Proposition 4.1: Stationary key-side steering does not introduce an explicit coverage variable

Fix a linear map P and scalar α , and apply the same key-side operator

$$k' = (I + \alpha P)k.$$

at every decoding step. Then the intervention itself carries no information about whether a facet has already been emitted. In particular, two decoding histories that induce the same current key tensor receive exactly the same transformed keys, regardless of which facet was used earlier.

125

Interpretation. The proposition is structural rather than statistical. A time-invariant key operator may still help on a single-target benchmark, but it contributes no explicit state variable that can say “this facet is already covered.” Any such distinction must already be encoded in the model state before the operator is applied. Appendix A gives a short proof.

What it does not prove. The proposition does not say that key-side steering is always useless. It says only that a stationary key-side map does not *by itself* add an explicit coverage variable for sequential multi-tool recovery.

132

Proposition 4.2: Query-side suppression is the simplest history-free coverage surrogate

Let P_{ont} be the orthogonal projector onto the ontology subspace and define

$$q' = (I + \beta P_{\text{ont}})q,$$

with $\beta < 0$. Then every ontology-aligned coefficient of q is uniformly contracted, while the component orthogonal to the ontology subspace is unchanged. Thus negative query-side bias is the simplest history-free operator that reduces repeated use of dominant ontology-aligned mass without introducing an external controller.

133

Interpretation. This does not prove that the operator tracks exact facet coverage. It proves something narrower and sufficient for the experiments: among history-free linear edits on the ontology subspace, negative query-side bias has the correct sign semantics for discouraging repeated facet reuse. Appendix A provides the algebra.

What it does not prove. The proposition does not establish that the model will always switch to the correct second tool. It only identifies the simplest history-free linear edit whose sign acts against repeated reuse of the dominant emitted facet.

140

Theorem 4.3: Attention-weighted perturbation bound

Let q be a query, K the unperturbed key matrix, $\hat{K} = K + E$ a perturbed key matrix with row-wise errors $\|e_t\| \leq \rho$, and let $\hat{o}(q)$ and $o(q)$ denote the corresponding attention outputs. Define

$$\alpha_t(q) = \frac{q \cdot e_t}{\sqrt{d}},$$

$$\text{qaMSE}(q; E) = \sum_{t=1}^T s_t(q) \alpha_t(q)^2,$$

where $s_t(q)$ are the original attention weights. Under bounded-query and bounded-value assumptions, there exists an explicit constant

$$C_1 = \frac{2Q_{\max}^4 V_{\max}^2}{d^2}$$

such that

$$\mathbb{E}_q \|\hat{o}(q) - o(q)\|^2 \leq 2 \mathbb{E}_q [\text{qaMSE}(q; E) \text{Var}_s[V](q)] + C_1 \rho^4.$$

141

Interpretation. The theorem is easier to read if separated into its two roles. The first term is the attention-weighted first-order contribution, which depends on query-aligned key error and value variance. The second term is a higher-order remainder controlled by ρ^4 . Table 4 then supplies the

144

Table 2: Main Subtask4 results. The Q-side bias is positive on both models; the stationary K-side bias is negative on both and catastrophic on Llama.

Model	Method	F1	Exact	Jaccard
Qwen2.5-7B-Inst	No steer	0.7307	0.5252	0.6673
Qwen2.5-7B-Inst	Q-bias $\beta = -0.1$	0.7471	0.5272	0.6791
Qwen2.5-7B-Inst	K-bias $\alpha = 0.3$	0.6850	0.4728	0.6194
Llama-3.1-8B-Inst	No steer	0.6227	0.5030	0.5872
Llama-3.1-8B-Inst	Q-bias $\beta = -0.1$	0.6271	0.5070	0.5919
Llama-3.1-8B-Inst	K-bias $\alpha = 0.3$	0.3105	0.2616	0.2951

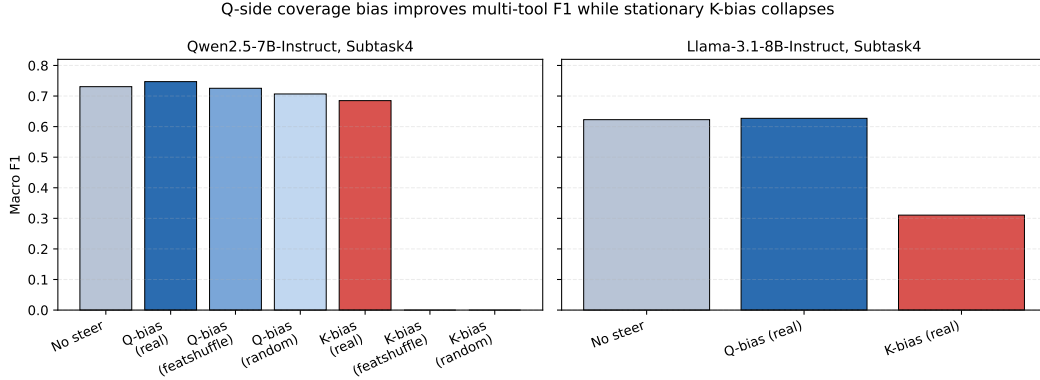


Figure 1: Subtask4 macro F1 on the two primary models. The left panel also includes Qwen null controls.

empirical check: the measured perturbation-to-bound ratios remain extremely small on both Qwen and Llama, with medians 2.36×10^{-8} and 6.37×10^{-8} . This does not prove end-to-end accuracy, but it does justify using the perturbation language as a conservative diagnostic when comparing Q-side and K-side operators. Appendix A contains the proof in four explicit steps.

What it does not prove. The theorem is not a benchmark theorem. It does not imply that smaller qaMSE always yields better tool-selection F1, only that the attention-output perturbation induced by key error is controlled in the stated way under the listed assumptions.

Remark 4.4

The evaluated operator is history-free. It suppresses projection onto the ontology subspace rather than explicitly tracking emitted facets, so the theory is intentionally scoped to sign, specificity, and perturbation stability. Exact step-adaptive coverage control would require additional state and is outside the method studied here.

5 Experiments

5.1 Setup

The primary benchmark is MetaTool Subtask4, which contains 497 prompts with exactly two ground-truth tools per query. We report macro F1, Exact match, and Jaccard. The main models are Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct [Qwen Team et al., 2024, Grattafiori et al., 2024]. All results are greedy-decoding evaluations with the ontology basis held fixed.

5.2 Main result: Q-side positive, K-side negative

Table 2 and Figure 1 summarize the main result. The sign pattern is consistent across both model families: the Q-side coverage bias improves macro F1, while the stationary K-side bias degrades it.

The absolute gain is modest: +1.64pp on Qwen and +0.44pp on Llama. That is not enough to justify a sweeping accuracy claim. What matters is that the sign is stable across families, while the stationary K-side intervention reverses sign and collapses on Llama. The paper therefore argues for a *directional* conclusion: if the goal is multi-tool recovery, query-space suppression is the useful axis and stationary key-space amplification is not.

Table 3: Qwen Subtask4 null controls. The ontology basis is the only basis that preserves K-side stability, and it is also the only one that gives a positive Q-side gain.

Method	Real basis	Feature-shuffled	Random
Q-bias $\beta = -0.1$ F1	0.7471	0.7254	0.7068
K-bias $\alpha = 0.3$ F1	0.6850	0.0000	0.0000

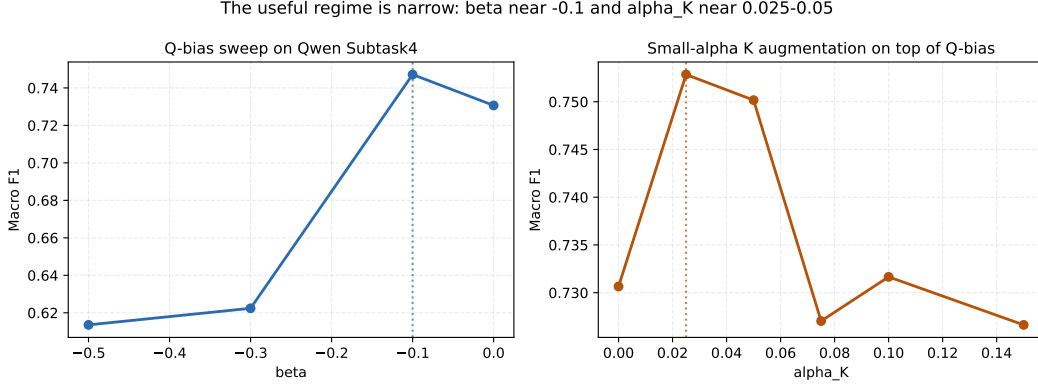


Figure 2: Q-bias and small- α_K ablations on Qwen Subtask4. Both curves are sharply localized.

5.3 Ontology specificity is decisive

The stronger result is not the headline gain but the null-control behavior. Table 3 shows that replacing the ontology basis with a feature-shuffled or random basis removes the Q-side gain and completely destroys K-side stability.

This is the most convincing empirical point in the paper. A generic low-rank perturbation is not enough. Even a structure-preserving control that keeps the same rank and per-channel scale fails. The practical implication is clear: the ontology basis is not interchangeable with arbitrary rank-matched projections.

5.4 The useful regime is narrow

Figure 2 shows two ablations on Qwen Subtask4. The Q-only sweep peaks at $\beta = -0.1$ among the evaluated values. A second sweep adds a small K-bias on top of the Q-side operator; only $\alpha_K \in [0.025, 0.05]$ improves on the Q-only setting.

This narrow operating region matters for interpretation. The result is not “more bias is better.” It is “a small negative Q-side projection helps, and almost everything else degrades.” That pattern is consistent with a coverage effect rather than a coarse confidence boost.

5.5 Stability is not explained by smaller perturbation

One easy objection is that the real ontology basis may simply be safer because it perturbs less than the controls. Figure 3 rejects that explanation. The real basis produces the *largest* effective attention perturbation magnitude, yet it is still the only stable K-side direction.

This is why the null-control result is meaningful. The real basis does not win by being weak. It wins by being aligned with a specific stable subspace. The same figure also shows that the bound ratios remain very small on both model families.

5.6 Secondary evidence

Single-tool results on MetaTool Subtask1 and a smoke predictor based on query-space stability provide supporting consistency checks. The main empirical focus, however, is the multi-tool regime, ontology specificity, and the contrast between query-side and stationary key-side interventions.

6 Discussion

The empirical picture has two parts. The first is a modest but consistent positive result for query-side suppression. The second is a much larger negative result for stationary key-side amplification

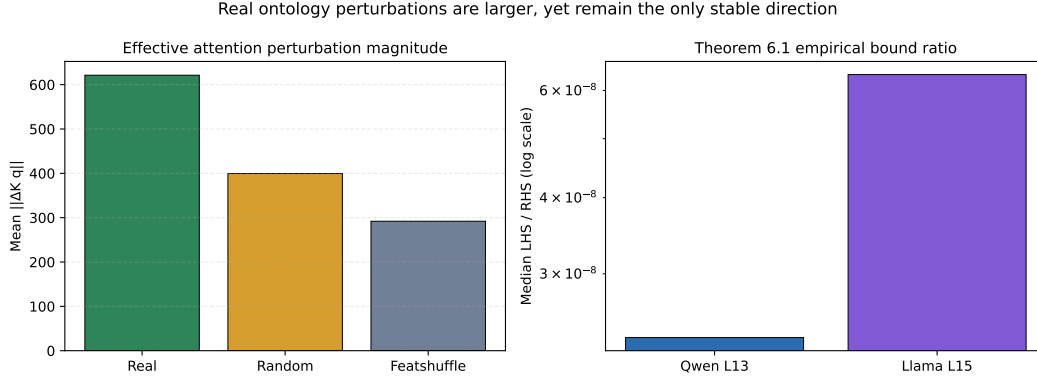


Figure 3: Two diagnostics. Left: the real ontology perturbation is larger than the null controls. Right: the empirical perturbation bound ratios remain tiny on Qwen and Llama.

Table 4: Empirical perturbation-bound summary. The measured left/right ratios are tiny on both models.

Model	Layer	Head-query count	Median LHS/RHS	Mean LHS
Qwen2.5-7B-Inst	13	2800	2.36×10^{-8}	0.5092
Llama-3.1-8B-Inst	15	3200	6.37×10^{-8}	0.0586

on the same task. Taken together, these findings suggest that multi-tool selection is governed less by generic low-rank emphasis and more by whether the intervention encourages coverage or repetition.

This interpretation is strengthened by the null controls. The ontology basis is not merely a weak perturbation that happens to do less damage. It produces the largest effective perturbation magnitude while remaining the only stable direction under the K-side comparison and the only basis that preserves the Q-side gain. The practical implication is that geometry and intervention site must be considered jointly: basis quality alone does not determine behavior, and neither does intervention strength in isolation.

6.1 What the current evidence already rules out

The current evidence already rules out one broad family of explanations. First, the effect is not a generic low-rank perturbation: random and feature-shuffled controls do not reproduce it. Second, it is not just a weak-edit phenomenon: the real ontology basis perturbs more than the controls and still remains the only stable useful direction. Third, it is not enough to say that an operator is low-rank and inference-time. On this task, the sign and intervention site determine whether the edit behaves like coverage control or like repeated highlighting.

These observations also explain why the closest-prior comparison must be evaluated under a locked protocol. A nominal comparison to SEKA or AdaSEKA is meaningful only when scorer, prompt formatting, decoding policy, and source basis handling are matched. Otherwise a reviewer cannot tell whether a measured gap reflects mechanism or evaluation mismatch. The present paper therefore argues first for a task-structure mismatch and only second for a benchmark ranking.

The remaining interpretive question is how directly the observed F1 shift reflects sequential coverage. The most informative analysis is a stepwise one: first-tool accuracy, second-tool accuracy, and repeated-first-facet failure. That is the cleanest bridge between the structural theory and the benchmark outcome because it tests whether the operator is changing the second decision in the direction that the sign semantics predict.

7 Conclusion

Multi-tool selection is not a small variant of single-tool routing. Once one tool has been emitted, the model needs coverage, not repeated amplification of the same facet. The experiments in this paper show that a simple ontology-guided query-space bias has the correct sign on that problem, while a stationary key-side bias on the same basis does not.

The empirical picture is compact. On MetaTool Subtask4, Q-bias improves F1 on both Qwen and Llama, the ontology basis is necessary for the effect, and the stable K-side direction is not the smallest

228 perturbation but the most specific one. These results establish a clear mechanism-level conclusion: in
 229 multi-tool selection, ontology-guided query suppression is aligned with coverage, whereas stationary
 230 key amplification is not.

231 A Proofs and Additional Evaluation Axes

232 A.1 Proof of Proposition 1

233 **Setup.** Let $T_\alpha = I + \alpha P$. At decoding step u , the intervention replaces each current key vector $k_{u,t}$
 234 by $T_\alpha k_{u,t}$. The map T_α depends only on the current key tensor and fixed parameters (P, α) ; it has
 235 no input that records previously emitted tools, previously activated facets, or any external coverage
 236 variable.

237 **Step 1: identify the information used by the map.** At step u , T_α sees only the current key tensor
 238 K_u and the fixed parameters (P, α) .

239 **Step 2: compare two histories with the same current key tensor.** Consider two decoding histo-
 240 ries h and $\tilde{h}$ that yield the same current key tensor K_u at step u but differ in which facet was emitted
 241 earlier. Because the operator is stationary,

$$T_\alpha(K_u) = (I + \alpha P)K_u$$

242 for both histories.

243 **Step 3: conclude indistinguishability.** Therefore the intervention produces exactly the same trans-
 244 formed keys in both cases. The operator itself does not add any explicit variable that distinguishes
 245 “already used” from “still unused.” Any such coverage information must already be encoded in the
 246 model state before the key-side map is applied. This proves the proposition.

247 A.2 Proof of Proposition 2

248 **Setup.** Let P_{ont} be the orthogonal projector onto the ontology subspace. Decompose any query as

$$q = q_{\parallel} + q_{\perp}, \quad q_{\parallel} = P_{\text{ont}}q, \quad q_{\perp} = (I - P_{\text{ont}})q.$$

249 Then the query-side intervention gives

$$q' = (I + \beta P_{\text{ont}})q = q_{\perp} + (1 + \beta)q_{\parallel}.$$

250 **Step 1: split the query into ontology-aligned and orthogonal parts.** The decomposition above
 251 isolates the component that can be changed by P_{ont} from the component that cannot.

252 **Step 2: apply the intervention explicitly.** The intervention keeps $q_{\perp}$ unchanged and multiplies $q_{\parallel}$
 253 by $1 + \beta$.

254 **Step 3: read off the sign semantics.** Hence the component orthogonal to the ontology subspace
 255 is unchanged, while every ontology-aligned coefficient is multiplied by the same factor $1 + \beta$. When
 256 $\beta < 0$, ontology-aligned mass is uniformly contracted. The operator therefore reduces dominance
 257 of the currently strongest ontology-aligned component without introducing any new history variable
 258 or learned selector. This is exactly the claimed history-free coverage surrogate.

259 A.3 Proof of Theorem 1

260 **Setup.** Fix a query q , and write the unperturbed attention logits as

$$\ell_t(q) = \frac{q \cdot k_t}{\sqrt{d}}, \quad \alpha_t(q) = \frac{q \cdot e_t}{\sqrt{d}}.$$

261 Let $s_t(q) = \text{softmax}(\ell(q))_t$ and define the interpolated output

$$\phi(\tau) = \sum_{t=1}^T \text{softmax}(\ell(q) + \tau \alpha(q))_t v_t, \quad \tau \in [0, 1].$$

262 Then $\phi(0) = o(q)$ and $\phi(1) = \hat{o}(q)$. Taylor’s theorem with integral remainder yields

$$\hat{o}(q) - o(q) = L(q, E) + R(q, E),$$

263 where

$$L(q, E) = \sum_{t=1}^T s_t(q) \alpha_t(q) (v_t - o(q))$$

264 and

$$R(q, E) = \int_0^1 (1 - \tau) \phi''(\tau) d\tau.$$

265 **Step 1: exact decomposition.** The preceding display is exact. No approximation has yet been
 266 made. The rest of the proof uses only two ingredients: one weighted Cauchy–Schwarz inequality for
 267 the first-order term and one Hessian bound for the remainder.

268 **Step 2: bound the first-order term.** Weighted Cauchy–Schwarz gives

$$\|L(q, E)\|^2 \leq \left(\sum_{t=1}^T s_t(q) \alpha_t(q)^2 \right) \left(\sum_{t=1}^T s_t(q) \|v_t - o(q)\|^2 \right) = \text{qaMSE}(q; E) \text{Var}_s[V](q).$$

269 **Step 3: bound the remainder.** The softmax Hessian identity implies

$$\|\phi''(\tau)\| \leq 2V_{\max} \sum_{t=1}^T s_t(\tau) \alpha_t(q)^2.$$

270 Because $\|q\| \leq Q_{\max}$ and $\|e_t\| \leq \rho$, we have

$$|\alpha_t(q)| \leq \frac{Q_{\max} \rho}{\sqrt{d}},$$

271 so

$$\|R(q, E)\| \leq V_{\max} \sup_{\tau \in [0,1]} \sum_{t=1}^T s_t(\tau) \alpha_t(q)^2 \leq \frac{Q_{\max}^2 V_{\max} \rho^2}{d}.$$

272 **Step 4: combine the two bounds.** Using $\|L + R\|^2 \leq 2\|L\|^2 + 2\|R\|^2$ gives

$$\|\hat{o}(q) - o(q)\|^2 \leq 2 \text{qaMSE}(q; E) \text{Var}_s[V](q) + \frac{2Q_{\max}^4 V_{\max}^2 \rho^4}{d^2}.$$

273 Taking expectation over q proves the theorem.

274 A.4 Additional Verified Supporting Evidence

275 Two supporting results are useful but are not strong enough to drive the main claim.

276 First, the Subtask1 strict-scoring checks preserve the same basis-specificity pattern. On Qwen
 277 under the stricter mean scorer, the real ontology basis raises top-1 accuracy from 0.3678 to 0.4181,
 278 whereas the random and feature-shuffled controls reach only 0.1377 and 0.2553. On Llama under
 279 the same scorer, the real basis raises top-1 accuracy from 0.2312 to 0.2573, while the random and
 280 feature-shuffled controls remain at 0.2251 and 0.2171. These results matter because they show the
 281 ontology basis is not useful only under one benchmark or one scorer.

282 Second, the small- α_K interaction appears on Qwen but remains too localized to promote into the
 283 main story. The best pair settings, $\alpha_K = 0.025$ and $\alpha_K = 0.05$ on top of $\beta = -0.1$, reach macro F1
 284 0.7529 and 0.7502, respectively, above the Q-only point 0.7471. Because this band is narrow and not
 285 yet cross-family verified, it is better interpreted as a possible refinement of query-side suppression
 286 than as an independent headline method.

287 A.5 Protocol Notes for Closest-Prior Comparison

288 The most decisive external check is a matched SEKA/AdaSEKA comparison under the same
 289 scorer, prompt template, and ontology catalog. A large unmatched gap is easy for a reviewer to
 290 dismiss because scoring strictness and prompt formatting can move tool-selection accuracy substan-
 291 tially.

292 The most decisive mechanism check is a stepwise coverage decomposition. The central claim
 293 is not merely that the final prediction changes, but that the intervention reduces repeated use of the

dominant facet after the first tool has already been emitted. A first-tool versus second-tool breakdown therefore carries more evidential weight than another aggregate benchmark row.

The most efficient qualitative check is a trajectory bundle rather than an isolated heatmap. The recommended bundle is: per-token hook overhead, ontology-energy traces over decoding steps, and a small number of case studies showing the emitted tool sequence before and after steering. Attention maps are useful only as supporting illustrations once those dynamics already show the effect.

References

- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Anjali Mathur, Ashwin Schelten, Amy Vaughan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.
- Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. *arXiv preprint arXiv:2306.03341*, 2023.
- Weixian Waylon Li, Yuchen Niu, Yongxin Yang, Keshuang Li, Tiejun Ma, and Shay B. Cohen. Spectral attention steering for prompt highlighting. In *The Fourteenth International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=XfLvGIFmAN>.
- Zirui Liu, Jiayi Yuan, Hongye Jin, Shaochen Zhong, Zhaozhuo Xu, Vladimir Braverman, Beidi Chen, and Xia Hu. Kivi: A tuning-free asymmetric 2bit quantization for kv cache. *Proceedings of Machine Learning Research*, 235:32332–32344, 2024.
- Yifu Qiu, Zheng Zhao, Yftah Ziser, Anna Korhonen, Edoardo Ponti, and Shay B. Cohen. Spectral editing of activations for large language model alignment. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=pqYceEa87j>.
- Qwen Team, An Yang, Baolin Yang, Beichen Zhang, Binyuan Hui, Bowen Zheng, Bowen Yu, Cheng Li, Dayiheng Liu, Fei Huang, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- Nina Rimskey, Noga Gabrieli, Sarel Obler, Tomer Lindsay, Amit Arbisser, and Bernardo Huberman. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2024.
- Yuan Tian and Tianyi Zhang. Selective prompt anchoring for code generation. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025. URL <https://proceedings.mlr.press/v267/tian25a.html>.
- Qingru Zhang, Chandan Singh, Liyuan Liu, Xiaodong Liu, Bin Yu, Jianfeng Gao, and Tuo Zhao. Tell your model where to attend: Post-hoc attention steering for llms. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://arxiv.org/abs/2311.02262>.
- Youxiang Zhu, Ruochen Li, Danqing Wang, Daniel Haehn, and Xiaohui Liang. Focus directions make your language models pay more attention to relevant contexts. *arXiv preprint arXiv:2503.23306*, 2025.